Sherwin Wang

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Education

Master of Computer Science **Dec 2026**
University of California, Irvine – Irvine, CA

Relevant Coursework: AI, Machine Learning, Neural Networks, Algorithms and Data Structures

B.S. Computer Science **May 2025**
Arizona State University – Tempe, AZ

Relevant Coursework: Computer Architecture, Operating Systems, Database Management

Technical Skills

Languages: Python, Java, C++, C#, SQL, JavaScript, HTML/CSS, Metal Shading Language
Frameworks & Libraries: PyTorch, MLX, React, Flask, Hugging Face, OpenCV, Tesseract, REST APIs
Tools & Platforms: Git, Docker, Linux, SQLite, Firebase, Shopify API, macOS (Apple Silicon), gem5

Experience

Software Developer Intern **May 2024 – Sept 2026**
Analytichem Canada (part-time / contract), Remote

- Migrated Analytichem's catalog of 3,000+ reference standards and reagents from SyteLine, the internal ERP, onto Shopify by writing SQL and Python that cleaned records into the CSV files Shopify's bulk import expects, then wrote scheduled Python jobs that pushed inventory, pricing, and CoA deltas to Shopify through its REST API. Two years live, over $500K in online sales.
- Built dynamic product search widgets for three product-line sites (pyodbc, SQL Server, SQLite/JSON on IIS) so search data could be updated in place without rebuilding the sites.
- Replaced 8+ legacy scripts across C#, Python, Access, and Excel for the Chemservice line with a single Python package that generates product HTML and Shopify import workbooks, diffs source data against the live Shopify catalog, and only writes safe additions. Hundreds of unit tests on the business rules.
- Built an internal admin app on a SQL backend hosted on Analytichem's Windows servers that let sales and operations staff edit listings, swap CoA files, and trigger updates. Every update ran dry-run and backup steps first so a bad input could not overwrite historical records.

Projects

Efficient LLM Inference on Apple Silicon (UCI Capstone) **Jan 2026 – May 2026**
Python, MLX, Metal Shading Language

- Wrote a custom Metal flash-attention GPU kernel that keeps the running softmax in on-chip memory instead of writing it out to GPU memory, and fused 10 GPU operations into 3 across the projection steps.
- Took generation from 40.9 to 74.0 tokens per second, a 1.81x speedup. Layered speculative decoding on top (small draft model, large target model, accept/reject in one forward pass) for another 1.22x on long outputs.
- Wrote the test harness that checked each kernel against a reference before trusting any speed numbers, so a wrong kernel could not pass as a fast one. Presented at the UCI capstone expo.

Food Pharmer: AI Nutrition Label Analyzer **Jan 2026 – Mar 2026**
Python, PyTorch, Tesseract, OpenCV

- Owned the OCR layer of a 5-person app that reads phone photos of nutrition labels and scores product healthiness. Benchmarked four OCR strategies and picked Tesseract, which hit 99.1% field accuracy on clean labels but dropped to 58.9% on real photos with glare and curved packaging.
- Trained an MLP (FieldConfidenceNet-Lite) on TF-IDF features of the raw OCR text plus the parsed value, unit, and a plausible-range flag, which predicted per-field extraction correctness at 0.972 AUROC. The app now trusts fields above 0.85 confidence and asks the user to retake photos for the rest.

Mixture-of-Experts Routing Strategy Comparison **Sep 2025 – Dec 2025**
Python, PyTorch, Hugging Face

- Trained four small transformers in PyTorch, identical except for how they routed tokens to expert layers, all at the same parameter count. The most evenly balanced router was the worst model of the four, about 25% below the winner, because its per-expert token limit was silently dropping tokens instead of processing them.